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Unit-IV

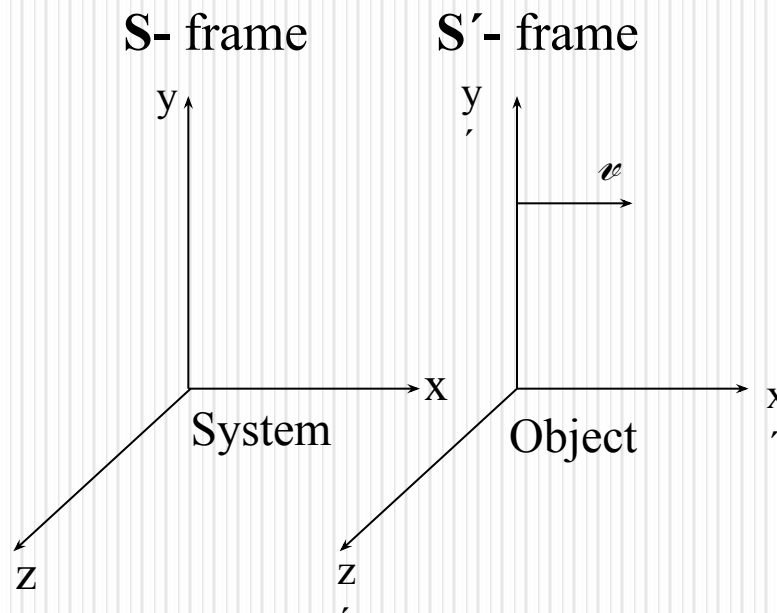
Theory of Relativity

Contents

- Introduction, Frame of reference, Galilean transformation eqs,
- Michelson-Morley experiment, 1887
- Postulates of special theory of relativity,
- Shortcomings in Galilean transformation
- Lorentz transformations equations 1895
- Length contraction, Time dilation, Experimental verification of time dilation, Relativistic addition of velocities, Simultaneity of two events, Variation of mass with velocity

Any Moving body (e.g. rocket, ball or atom etc.) - **Object**

A reference point w.r.t. which the object is moving - **System**



Object's frame is a moving frame as S'- frame

System's frame is a stationary frame as S- frame

Theory of relativity is divided into

Special theory of relativity

- deals with Object & System which are either moving at constant speed or at rest w.r.t. one another.
- Newton's laws are valid

General theory of relativity

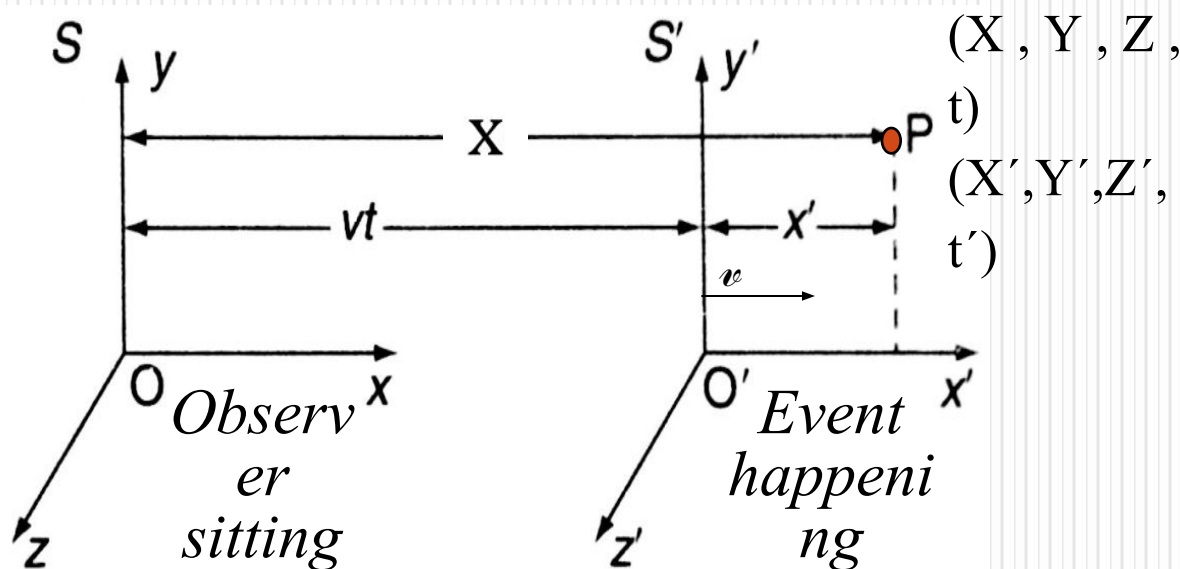
- deals with Object & System which are accelerated or decelerated w.r.t. one another.
- Newton's laws are not valid

Inertial frame of reference Non-Inertial frame of reference

A reference frame in which a particle obeys Newton's law of inertia i.e. it remains in its state of rest or moving with constant velocity if no force acts on it.

A reference frame in which a particle appears to be either accelerating or decelerating without a net force acting on it. A particle observed from this frame will not obey Newton's first law

Transformation equations transforms the measurements of space & time from one inertial frame of reference to another.



Let an event is happening in S' frame at point P at any time t' . Let space and time coordinates of event w.r.t to an observer in S & S' frame are (x, y, z, t) & (x', y', z', t')

respectively.

Galilean transformation equations (in 1-Dimension)

In Newtonian mechanics ($v \ll c$), objects are moving with very small velocities compared to velocity of light, the space & time transformation equations for transformation from S to S' frame are known as Galilean transformation equations given in 1-D as

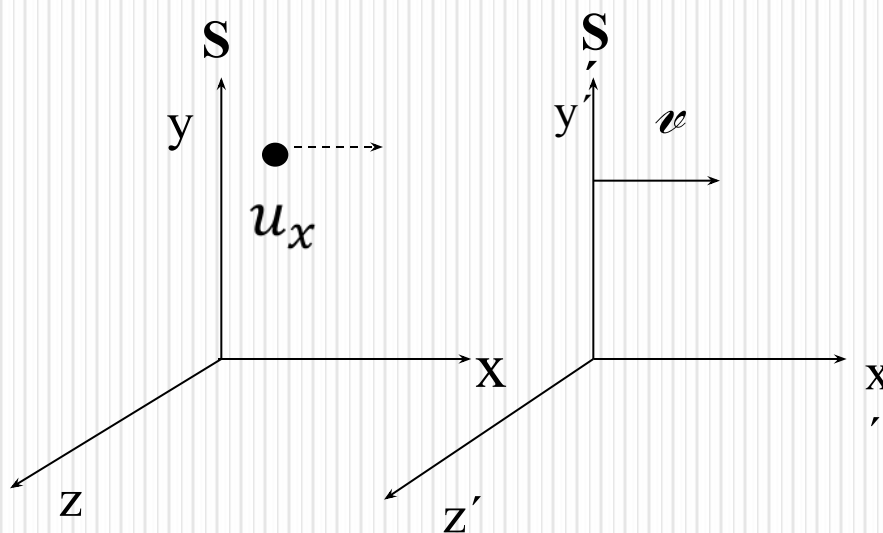
$$x' = x - vt \quad \text{----- (1)}$$

$$y' = y \quad \text{----- (2)}$$

$$z' = z \quad \text{----- (3)}$$

$$t' = t \quad \text{----- (4)}$$

Under Galilean transformation,
Transformation of “Velocity” of a particle is not invariant



Proof:

If particle is moving with velocity u_x along $+x$ in S-frame,

Then, from Galilean transformation, its velocity in S' frame is

$$\frac{dx'}{dt} = \frac{dx}{dt} - v$$

$$\frac{dx'}{dt'} = \frac{dx}{dt} - v$$

Since $dt' = dt$

$$\Rightarrow u_x' = u_x - v \quad \text{----- (5)}$$

Galilean Invariance: some quantities like mass, length, time, acceleration..etc. of particle do not change under Galilean transformation. “Newton’s laws of motion are invariant under Galilean transformation”

Proof: $\frac{dx'}{dt'} = \frac{dx}{dt}$

$$a' = a$$

$$ma' = ma \quad \text{where } m' = m \text{ as long as } v \ll c$$

$$\Rightarrow F' = F \quad \text{----- (6)}$$

Galilean Inverse transformation equations in 1-D

The space & time coordinate transformation equations for transformation from S' to S frame are called Galilean Inverse T.Eqs.

$$x = x' + v't'$$

$$y = y'$$

$$z = z'$$

$$t = t'$$

Note:

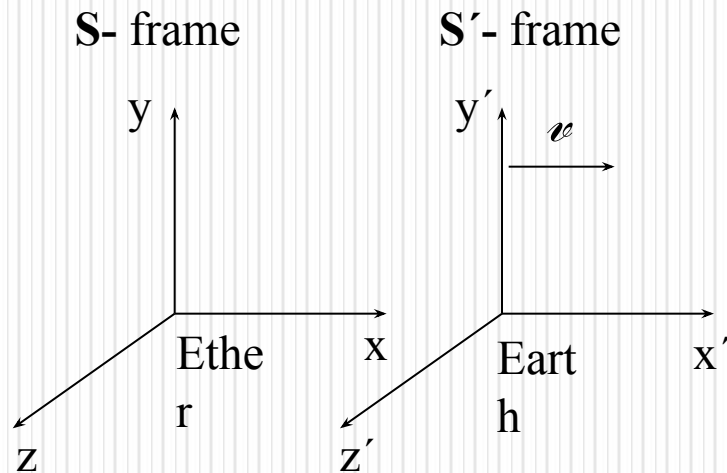
Galilean transformation for space & time are valid in Newtonian mechanics, i.e. when objects are moving with small velocities ($v \ll c$) compared to velocity of light.

Ether hypothesis for absolute frame

It was assumed that

“all bodies in the universe including our planet earth move freely through a hypothetical medium called ether”.

“Ether” was supposed to be a
massless,
invisible,
perfectly transparent &
perfectly non-resistive medium
at absolute rest.



If ether is at rest then for any material body moving through it will be a relative motion b/n the two. As earth moves around the Sun in its orbit with very high velocity v It can be assumed that earth passes through stationary ether with speed v , consequently the absolute motion of moving body can be measured.

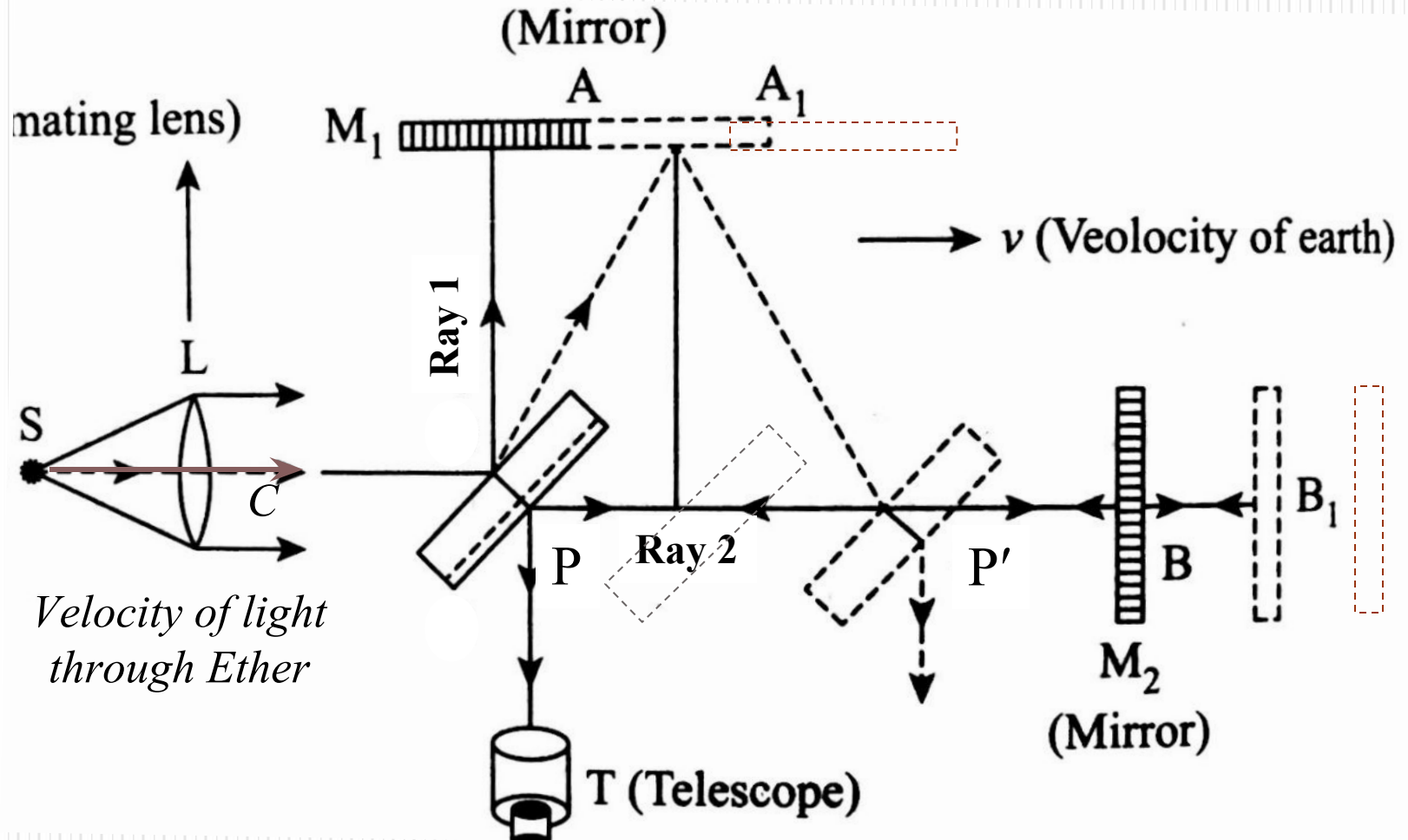
Michelson & Morley's Interferometer Experiment (1887)

Is there any existence of absolute frame of reference ?

Michelson & Morley performed an experiment to detect for any existence of absolute frame of reference. In their experiment they compared the speed of light in two perpendicular directions in an attempt to detect relative motion of earth through stationary ether.

Experiment: A monochromatic beam of light was incident on a semi silvered glass plate P, placed at 45° to incident beam. At P the ray is partly reflected & partly transmitted as

Ray 1 & Ray 2 respectively



Assuming that the direction of motion of earth coincides with the direction of incident beam of light.

- Following the hypothetical Ether , the whole experimental setup was moving w.r.t. the Ether (since the apparatus was placed on moving Earth), therefore it was expected that the two rays should take different time to return back to P.
- The motion of Earth through Ether can be determined by measuring the time difference 'dt' between two journeys (P to M_1 & back) & (P to M_2 & back) by Ray1 & Ray2 respectively.

In Transverse journey (P to M_1 & back)

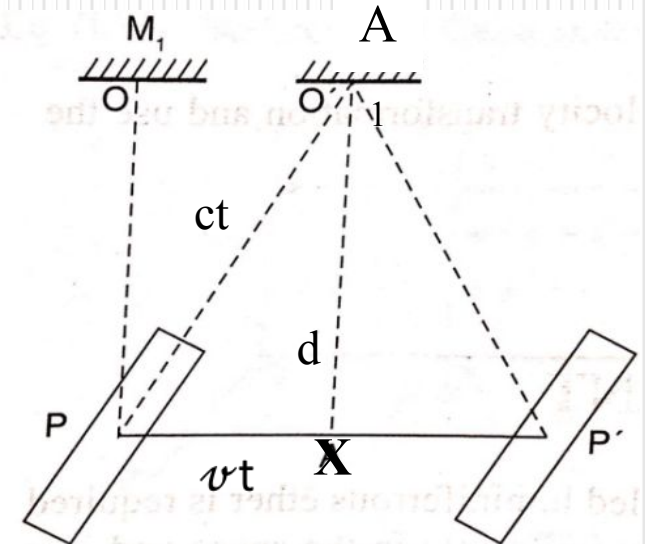
Ray 1 transversely strikes the mirror at A_1 & reflected back at P. Let 't' be the time taken in the journey from P to A_1

$$(ct)^2 = (vt)^2 + d^2$$

$$(c^2 - v^2)t^2 = d^2$$

$$t = \frac{d}{(c^2 - v^2)^{\frac{1}{2}}}$$

Since the velocity of light through Ether remains constant as C, the light will take same time in back journey from A_1 to P.



Hence the Total time taken by Ray1 in a complete transverse journey PA_1P' is

$$t_1 = 2t$$

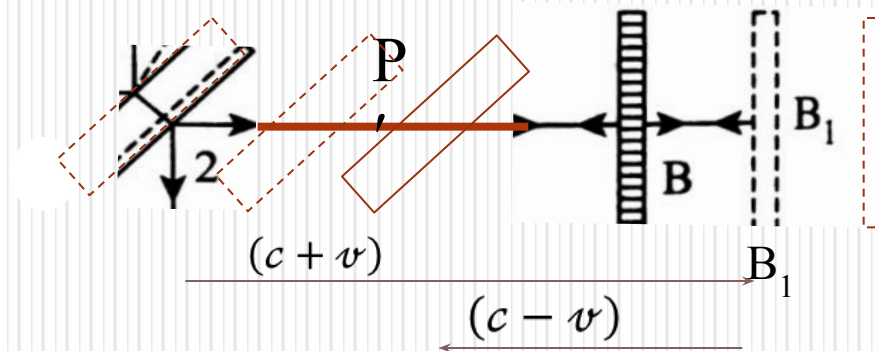
$$t_1 = \frac{2d}{(c^2 - v^2)^{\frac{1}{2}}}$$

$$t_1 = \frac{2d}{c} \left(1 - \frac{v^2}{c^2} \right)^{-\frac{1}{2}}$$

as $v \ll c$, applying Binomial theorem

$$t_1 \approx \frac{2d}{c} \left(1 + \frac{v^2}{2c^2} \right) \quad \text{----- (1)}$$

In Longitudinal journey (P to M_2 & back)



Total time taken by Ray 2 from P to B_1 and back P'

$$t_2 = \text{time taken for (P to } B_1) + \text{time taken for (} B_1 \text{ to P')}$$

$$t_2 = \frac{d}{(c-v)} + \frac{d}{(c+v)}$$

$$= \frac{2cd}{(c^2 - v^2)}$$

$$t_2 = \frac{2cd}{c^2} \left(1 - \frac{v^2}{c^2}\right)^{-1} \text{ as } v \ll c, \text{ applying Binomial theorem,}$$

$$t_2 \approx \frac{2d}{c} \left(1 + \frac{v^2}{c^2}\right) \text{ ----- (2)}$$

The time difference in two journeys

$$dt = t_2 - t_1$$

$$dt = \frac{2d}{c} \left(1 + \frac{v^2}{c^2}\right) - \frac{2d}{c} \left(1 + \frac{v^2}{2c^2}\right)$$

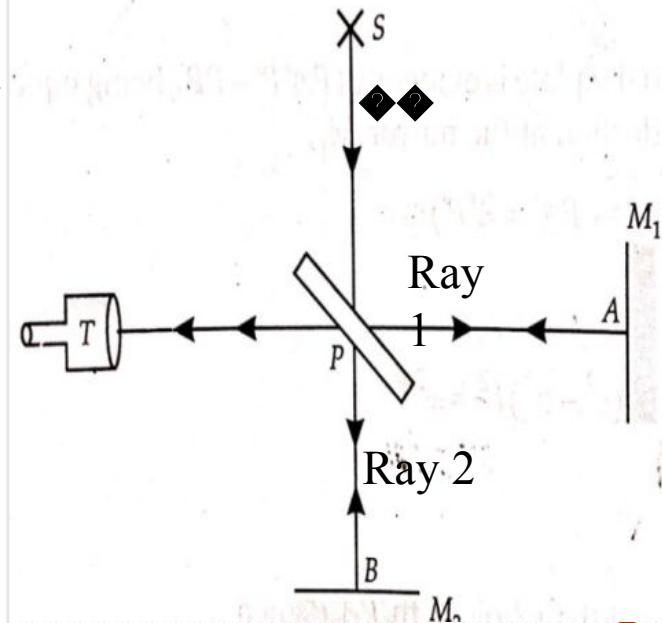
$$dt = \frac{dv^2}{c^3} \text{ ----- (3)}$$

This time difference introduces an optical path difference between two interfering beams as $\Delta = c \, dt$

$$\Delta = \frac{dv^2}{c^2} \quad \text{-----} \quad (4)$$

(Note: If there is any relative motion b/n ether & earth, some shift in the fringe system will be observed, otherwise there will not be any fringe shift)

However, to be able to notice any (if any) shift in the fringe due to moving earth, the whole apparatus was rotated by 90° . By doing so the path of two rays is interchanged.



Now the path difference becomes opposite w.r.t. previous arrangement.

$$\Delta_{\perp} = \frac{-d v^2}{c^2}$$

Resultant path difference

$$\delta = \Delta - \Delta_{\perp} = \frac{2d v^2}{c^2}$$

From the condition of constructive interference $\delta = N\lambda$,

A corresponding shift in the fringe system is given by

$$N = \frac{\delta}{\lambda} = \frac{2d v^2}{\lambda c^2}$$

According to Michelson, with the following data used in experiment

$$\lambda = 5500 \text{ \AA}$$

$$d = 11\text{m (by reflecting the beam back \& forth several times)}$$

$$v = 3 \times 10^4 \text{ m/s; } c = 3 \times 10^8 \text{ m/s;}$$

if there is any Earth's motion w.r.t. ether, the theoretical value of shift of $N=0.4$ fringes would be expected in fringe pattern on turning the apparatus by 90° . The Michelson-Morley's Experiment was sensitive enough to detect a fringe shift of the order of 0.01, however in spite of taking into account all considerations, no fringe shift was observe practically. The result was negative !!

Significance of the “Null result”

It suggests that there is no relative motion b/n Earth & Ether.
The motion of earth through “stationary ether” is a meaningless concept.

⇒ There is No absolute frame of reference.

⇒ The velocity of light is constant in all directions.

The failure of Michelson Morley experiment eventually led to the Postulates of Einstein's special theory of relativity

Postulates of Einstein's special theory of relativity

I. Absence of universal frame of reference

“The fundamental laws of Physics have same form in all inertial frames of reference”.

II. Constancy of speed of light “The speed of light in vacuum is a universal constant, it has the same value in all inertial frames of reference, regardless of their relative motion”.

Shortcomings of Galilean transformation for space & time

When particles are moving with very high velocity ($v \rightarrow c$) comparable to the velocity of light, they violate both the postulates of Special Theory of Relativity and break down.

Violation of Einstein's Postulates by Galilean transformation when ($v \rightarrow c$)

Violation of I postulate

Maxwell's equations in Electromagnetism are not remaining invariant under Galilean transformation.

Violation of II postulate

Speed of light is not remaining invariant under Galilean transformation; Violation of II postulate, e.g. If c is velocity of light in S-frame along x , then in S' frame the velocity is $c' = c - v$ which is different from that in S- frame.

This makes a requirement for new set of transformation equations.

Development of Lorentz transformation

1895

The theory of relativity is of great importance in microscopic world where velocities are enormously high ($v \rightarrow c$) e.g. for subatomic particles in nuclear interactions, for an electron accelerated through a P.D. of 10Mev (where speed equals $0.998c$).

For the microscopic world where particle's velocities are enormously high, Galilean transformation equations need to be modified. Lorentz developed the transformation equations for fast moving particles by directly using the Einstein's Postulates.

Lorentz's transformation equations for the transformation of coordinates from S to S' frame

$$x' = \frac{(x - vt)}{\sqrt{\left(1 - \frac{v^2}{c^2}\right)}} \quad \text{-----} \quad (1)$$

$$y' = y \quad \text{-----} \quad (2)$$

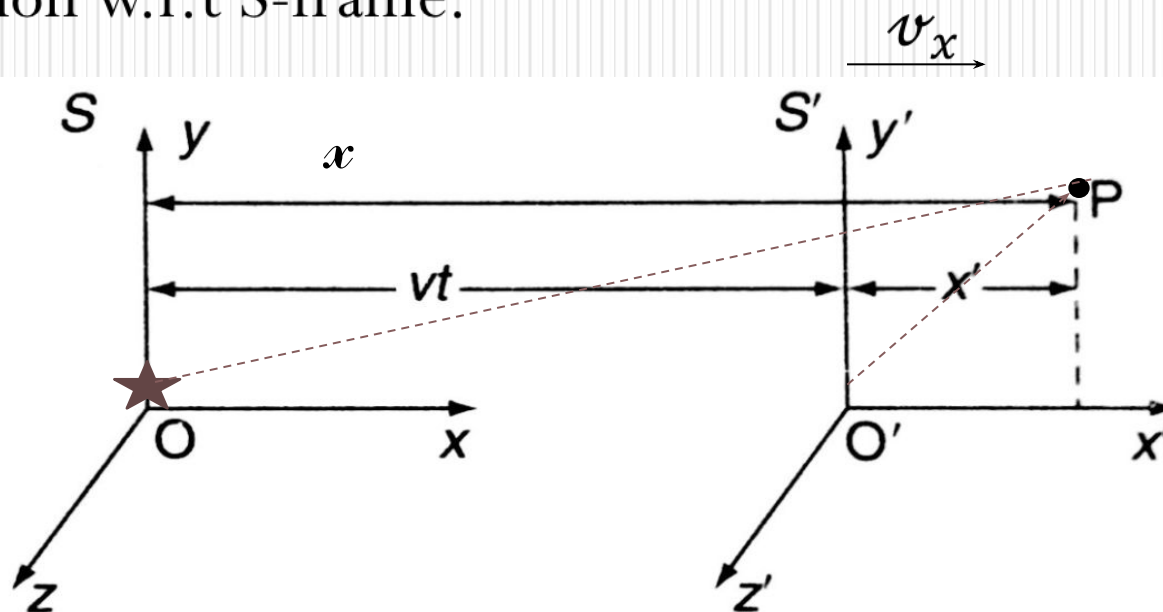
$$z' = z \quad \text{-----} \quad (3) \quad \text{and}$$

$$t' = \frac{\left(t - \frac{xv}{c^2}\right)}{\sqrt{\left(1 - \frac{v^2}{c^2}\right)}} \quad \text{-----} \quad (4)$$

Lorentz transformation (in 1-D) using Einstein's

Postulates

Let initially, when axis of two inertial frames S & S' frame are coinciding at $t=0$ & $t'=0$, a pulse of light is generated at the origin and frame S' starts moving with very high velocity ($v \rightarrow c$) along +ve x direction w.r.t S -frame.



Let at any later time, pulse of light reaches point P. The coordinates at pt. P w.r.t. S & S' frame respectively are (x, y, z, t) & (x', y', z', t')

Lorentz developed the transformation equations by directly using the Einstein's Postulates of Special theory of Relativity.

Let the transformation equation between x' & x is represented by

$$x' \propto (x - vt)$$

$$x' = k(x - vt) \quad \text{----- (1) in S' frame}$$

Let k is proportionality constant, independent of position & time.

- According to I- Postulate,

“the laws of Physics must have the same form in both S & S' frame”.

Therefore the corresponding equation of x in S frame must be

$$x = k(x' + vt')$$

----- (2) in S frame

$$y' = y$$

----- (3)

$$z' = z$$

----- (4)

However, the time coordinates t & t' are not equal, $t' \neq t$.

Substituting the value of x' from eq.(1) into eq. (2)

$$x = k[k(x - vt) + vt']$$

$$x = k^2x - k^2vt + kv t'$$

$$kv t' = x - k^2x + k^2vt$$

$$t' = \frac{x}{kv} (1 - k^2) + kt$$

$$t' = kt - \frac{x}{kv} (k^2 - 1)$$

On rearranging it gives

$$t' = kt - \frac{kx}{v} \left[1 - \frac{1}{k^2} \right] \quad \text{----- (5) in } S' \text{ frame}$$

Transformation Eqs. 1, 2, 3, 4 & 5 satisfies 1st postulate

• Following the II-Postulate to find the value of k,

“the constancy of speed of light in any frame of reference”.

Thus if pulse of light travels with same velocity i.e. c in both frames, then at later time its position in frame S & S' is given by

$$x = ct \quad \text{----- (6) in S frame}$$

$$x' = ct' \quad \text{----- (7) in S' frame}$$

Substituting the value of x & x' from eq.(6 & 7) into eq.(1)

$$ct' = k(ct - vt)$$

$$ct' = kt(c - v) \quad \text{----- (8)}$$

Similarly substituting the value from eq.(6 & 7) into eq.(2) gives

$$ct = kt'(c + v) \quad \text{----- (9)}$$

Multiplying eq. (8) & (9) gives the value of K

$$c^2 tt' = k^2 tt' (c^2 - v^2)$$

$$k^2 = \frac{c^2}{(c^2 - v^2)}$$

$$k = \frac{1}{\sqrt{\left(1 - \frac{v^2}{c^2}\right)}} \quad \text{----- (10)}$$

K is known as Lorentz's factor.

Knowing the value of k, equations (1, 3, 4 & 5) gives Lorentz's transformation equations, consistent with both postulates of Special theory of relativity.

Thus Lorentz's transformation equations for the transformation of coordinates from S to S' frame are as follows:

$$x' = k(x - vt) \quad \text{Or} \quad x' = \frac{(x - vt)}{\sqrt{\left(1 - \frac{v^2}{c^2}\right)}} \quad \text{-----} \quad (1)$$

$$y' = y \quad \text{-----} \quad (2)$$

$$z' = z \quad \text{-----} \quad (3)$$

and

$$t' = k\left(t - \frac{vx}{c^2}\right) \quad \text{Or} \quad t' = \frac{\left(t - \frac{vx}{c^2}\right)}{\sqrt{\left(1 - \frac{v^2}{c^2}\right)}} \quad \text{-----} \quad (4)$$

When objects are moving with small velocities ($v \ll c$)
Lorentz's transformation eqs. reduce to Galilean Transformation eqs.

For $v \ll c$; $\frac{v^2}{c^2} \rightarrow 0$

Lorentz's factor $k = \frac{1}{\sqrt{\left(1 - \frac{v^2}{c^2}\right)}}$ reduces to $k = 1$,

$$x' = (x - vt) \quad \text{----- (1)}$$

$$y' = y \quad \text{----- (2)}$$

$$z' = z \quad \text{----- (3) and}$$

$$t' = t \quad \text{----- (4)}$$

Lorentz Inverse Transformation equations (in 1D) for transformation of coordinates from S' to S frame are:

$$x = k(x' + vt')$$
----- (1)

$$y = y'$$
----- (2)

$$z = z'$$
----- (3) and

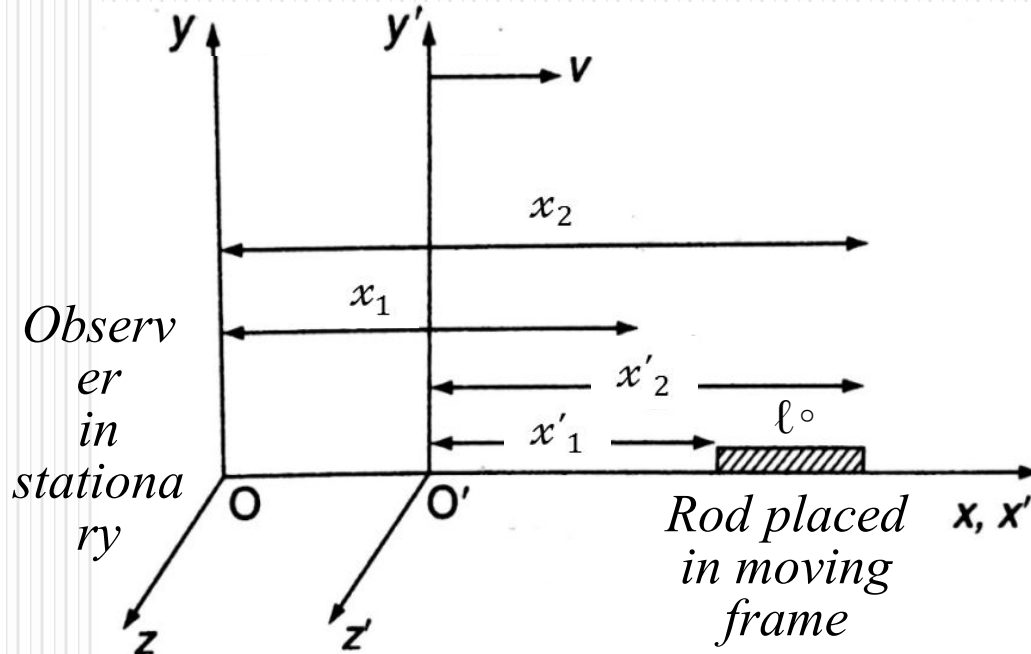
$$t = k\left(t' + \frac{x'v}{c^2}\right)$$
----- (4)

Where K is known as Lorentz's factor, $k = \frac{1}{\sqrt{\left(1 - \frac{v^2}{c^2}\right)}}$

Consequences of Lorentz transformations are propositions that

- Length Contraction,
objects appear to contract in the direction of motion
- Time dilation,
rate of a moving clock seems to decrease
- Relativistic addition of velocities,
maximum velocity attainable in the universe is that of light
- Simultaneity of two events
- Relativistic variation of Mass with velocity

Length contraction: When an object is moving with very high velocity comparable to the velocity of light, the length of moving object measured by a stationary observer, appears to be contracted (shorten) along the direction of motion. **This is length contraction.**



Consider rod of length ℓ_0 placed in moving frame S' , lying along x'

$$\ell_0 = x'_2 - x'_1 \quad \text{-----} \quad (1)$$

From Lorentz transformation eq.

$$x' = \frac{x - vt}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\ell_0 = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} [(x_2 - vt) - (x_1 - vt)]$$

$$\ell_0 = \frac{(x_2 - x_1)}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$x_2 - x_1 = \ell_0 \sqrt{1 - \frac{v^2}{c^2}}$$

Hence the length of rod as measured using Lorentz transformation eq by an observer in Lab frame (stationary frame) is given by

$$\ell = \ell_0 \sqrt{\left(1 - \frac{v^2}{c^2}\right)}$$

For very high velocities, $v \rightarrow c$, $v/c \rightarrow 1$; $1 - \left(\frac{v}{c}\right)^2 < 1$

$\ell < \ell_0$ Length of moving rod to a stationary observer in lab frame appears to be contracted in the direction of motion.

For small velocities, $v \ll c$, $\left(\frac{v}{c}\right)^2$ becomes negligible,

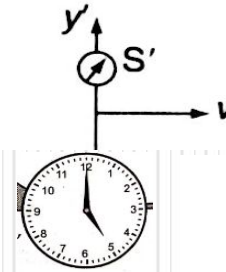
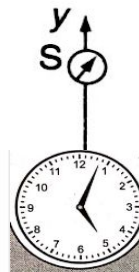
$\ell = \ell_0$ Length of moving rod remains unaffected.

$v = c$ and $v > c$; Impossible

Time dilation when $(v/c \rightarrow 1)$

Suppose an event happens in a spacecraft moving with very high velocity. Let t'_1 & t'_2 represent the time of start & end of event in S' frame. If t_1 & t_2 represents the corresponding time for the same event observed by a stationary observer from lab frame (S -frame), the time duration of an event Δt measured by a stationary observer appears to be lengthened than the actual or proper time interval of event. **This is Time dilation.**

Observe
 r
in
Stationa



Event
happening
in
moving frame

Time duration of event from Lab frame (stationary frame)

$$\Delta t = t_2 - t_1$$

From Lorentz inverse transformation eq. $t = \frac{t' + \frac{x'v}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$

$$\Delta t = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \left[\left(t'_2 + \frac{x'v}{c^2} \right) - \left(t'_1 + \frac{x'v}{c^2} \right) \right]$$

$$\Delta t = \frac{(t'_2 - t'_1)}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Where $\Delta t_o = (t'_2 - t'_1)$ represents the proper time i.e. the time duration of event in moving frame

For very high velocities, $v \rightarrow c$, $v/c \rightarrow 1$ $1 - \left(\frac{v}{c}\right)^2 < 1$

$\Delta t > \Delta t_0$ Time interval of an event appears to be lengthened to a stationary observer in Lab frame.

For small velocities, $v \ll c$, $\left(\frac{v}{c}\right)^2$ becomes negligible,

$\Delta t = \Delta t_0$ Time interval for an event remains unaffected in lab frame.

For $v = c$; $\Delta t = 0$; Impossible

For $v > c$; $\Delta t = \Delta t_0 \sqrt{-ve}$; meaningless, Impossible

Experimental verification of Time dilation

μ -meson particles are created by fast cosmic ray particles at a height of $\sim 10\text{kms}$ from the surface of earth. They move with Velocity $v = 2.99 \times 10^8 \text{ m/s} = 0.998c$. The average life is $\Delta t_0 = 2.0 \times 10^{-6} \text{ sec}$, after which they decay into an electron.

$$\mu \rightarrow e + \nu_0$$

This shows that in its life time it can travel a distance of $v \Delta t_0$

$$d' \cong 600\text{m} = 0.6\text{km!}$$

However these particles are observed near earth in large numbers.

How do μ -meson travel a distance of $\sim 10\text{kms}$ to reach the earth?

By Lorentz transformation, in Lab frame, the life time of μ -meson

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{2.0 \times 10^{-6}}{\sqrt{1 - (0.998)^2}}$$
$$\Delta t = 3.17 \times 10^{-5} \text{ sec.}$$

Thus a meson in its life time in Lab frame can travel a distance of

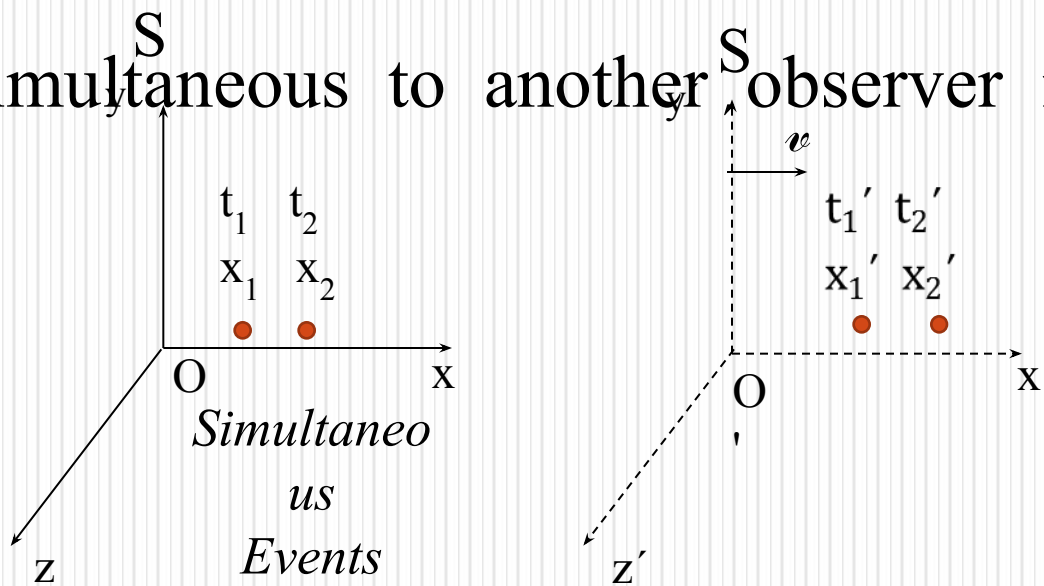
$$d = v\Delta t = 2.99 \times 10^8 \times 3.17 \times 10^{-5} = 9.5 \text{ km.}$$

By the effect of time dilation from Lorentz transformation, in Lab frame on Earth, the life time of μ -meson Δt , appears to be lengthened. Hence, despite their brief life time it is

possible for mesons to reach the Earth from high altitude

Simultaneity of two events (in 1-D)

Events that seems to take place simultaneously to one observer may not be simultaneous to another observer in relative motion



Proof as follows

In S-frame, let two events 1 & 2 are happening simultaneously at same time $t_1 = t_2$, but at different positions respectively.

Time coordinate from Lorentz Transformation : $t' = \frac{t - xv/c^2}{\sqrt{1 - \frac{v^2}{c^2}}}$

$$t_2' - t_1' = \left[\frac{t_2 - x_2 v/c^2}{\sqrt{1 - \frac{v^2}{c^2}}} \right] - \left[\frac{t_1 - x_1 v/c^2}{\sqrt{1 - \frac{v^2}{c^2}}} \right] \quad \text{where } t_1 = t_2$$

$$t_2' - t_1' = (x_2 - x_1) \frac{v/c^2}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Since $x_2 - x_1 \neq 0 \quad \therefore \quad t_2' - t_1' \neq 0$

This shows that the events which are simultaneous ($t_2 = t_1$) in S-frame are not simultaneous ($t_2' \neq t_1'$) in S'-frame. The time is relative and is different for observers in relative motion.

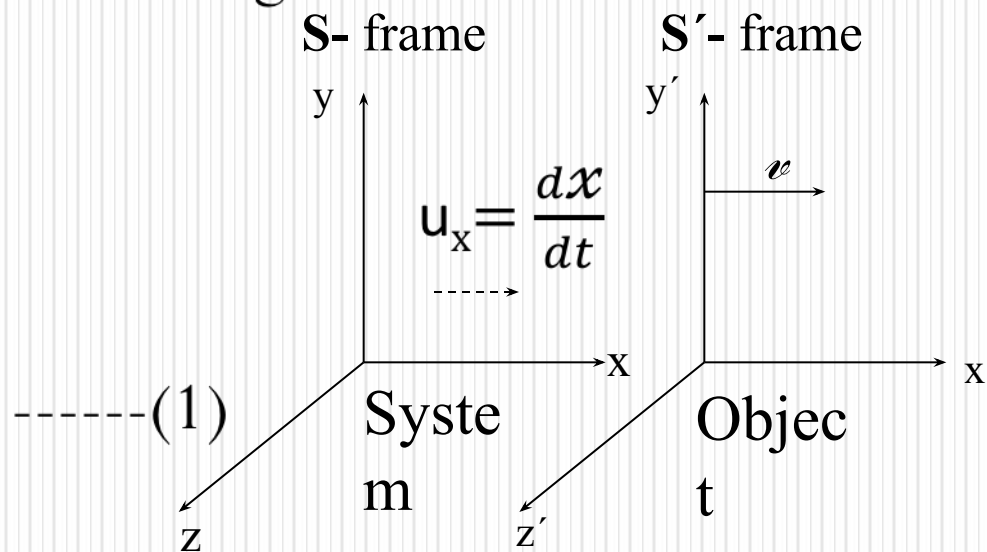
Relativistic addition of velocities (1-D)

When velocities are very high ($u, v \rightarrow c$), they are added relativistically consistent with Lorentz Transformation. They can not be directly added as in classical mechanics.

If in S-frame, a particle moves a distance $d\mathcal{x}$ in time dt , then from S'-frame it will appear to move through a distance $d\mathcal{x}'$ in time dt' .

$$u_x' = \frac{d\mathcal{x}'}{dt'} \times \frac{dt}{dt}$$

$$u_x' = \frac{d\mathcal{X}'/dt}{dt'/dt}$$



From Lorentz Transformation

$$x' = \frac{x - vt}{\sqrt{1 - \frac{v^2}{c^2}}} \quad \text{-----}(2)$$

$$\frac{dX'}{dt} = \frac{u_x - v}{\sqrt{1 - \frac{v^2}{c^2}}} \quad \text{-----}(4)$$

$$t' = \frac{t - xv/c^2}{\sqrt{1 - \frac{v^2}{c^2}}} \quad \text{-----}(3)$$

$$\frac{dt'}{dt} = \frac{1 - u_x v/c^2}{\sqrt{1 - \frac{v^2}{c^2}}} \quad \text{-----}(5)$$

Differentiating eq.2 & 3 w.r.t time & putting their result into eq.(1) gives the velocity of particle relative to S'-frame as

$$u_x' = \frac{u_x - v}{1 - \frac{u_x v}{c^2}} \quad \text{and Similarly in S-frame } u_x = \frac{u_x' + v}{1 + \frac{u_x' v}{c^2}}$$

Speed of light remains same in all inertial frames

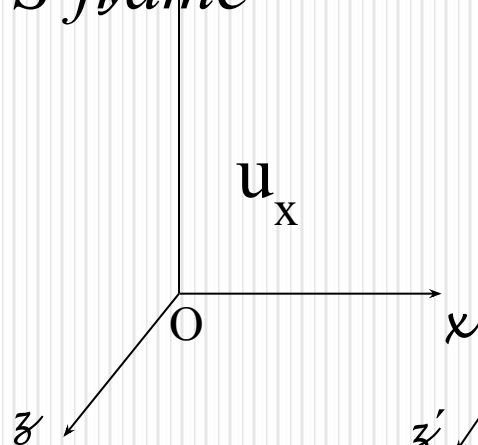
- If spaceship is moving with very high velocity ($v \rightarrow c$) along $+x'$ direction relative to Earth. If light signal is emitted from a spaceship along the direction of motion i.e. $u_x' = c$, then at any time t an observer in S-frame will record the same value for the velocity of light.

Proof: If $u_x' = c$,

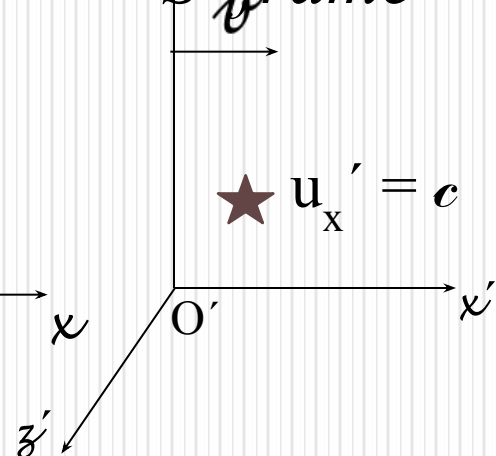
$$\text{then from } u_x = \frac{u_x' + v}{1 + \frac{u_x' v}{c^2}}$$

$$u_x = \frac{c + v}{1 + \frac{cv}{c^2}} \quad u_x = c$$

*Earth is
S-frame*



*Spaceship is
S'-frame*



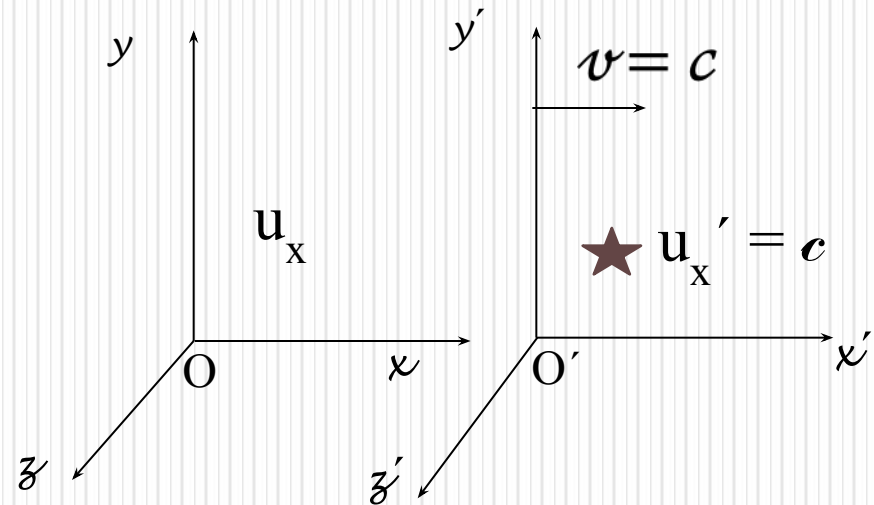
- Addition of velocity of light to the velocity of light reproduces the velocity of light.

Proof: If $u_x' = c$ and $v = c$,

then from $u_x = \frac{u_x' + v}{1 + \frac{u_x' v}{c^2}}$

$$u_x = \frac{c + c}{1 + \frac{cc}{c^2}}$$

$$u_x = c$$



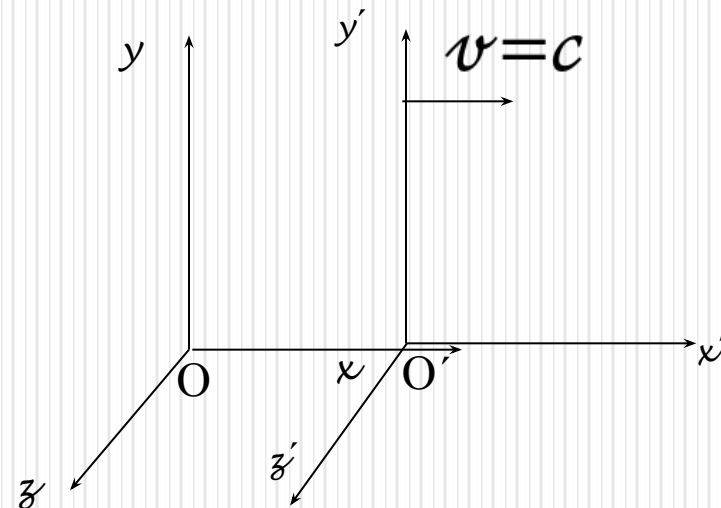
- If one object moves with velocity of light w.r.t. other i.e. $v = c$ then their relative velocity is always c , whatever may be the velocity of other.

Proof: If $v = c$ then

$$\text{from } u_x = \frac{u_x' + v}{1 + \frac{u_x' v}{c^2}}$$

$$u_x = \frac{u_x' + c}{1 + \frac{u_x' c}{c^2}}$$

$$u_x = c$$

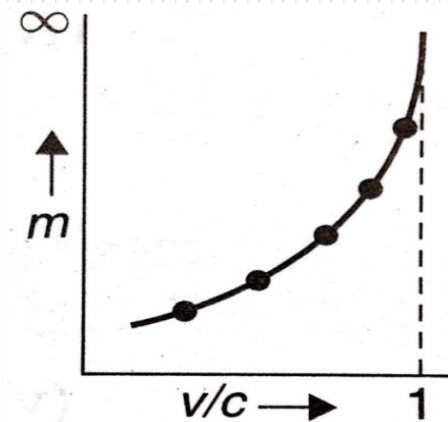


Relativistic variation of Mass with velocity

In Newtonian mechanics $v \ll c$, mass of a body is independent of velocity.

In Einstein's relativistic mechanics ($v \rightarrow c$) when particle is moving with very high velocity, the mass of a particle varies with velocity called the Relativistic Mass of a moving particle.

$$m(v) = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$



Where m_0 is the rest Mass of a particle (mass of particle at rest)

$$m > m_0$$

Relativistic mass of the particle is always greater than its rest mass.

- **For very high velocities, $v/c \rightarrow 1$ $1 - \left(\frac{v}{c}\right)^2 < 1$**
 $m > m_0$, Mass of a moving particle appears to be $>$ than its rest mass.
- **For small velocities, $v/c \ll 1$ $\left(\frac{v}{c}\right)^2$ becomes negligible**
 $m = m_0$ Relativistic mass reduces to rest mass
- When $v = c$; $m = \infty$; ;Impossible,
- When $v > c$; $m = m_0 \sqrt{-ve}$ meaningless, Impossible

Einstein's Mass energy equivalence (1905)

Before the theory of relativity, it was considered that mass and energy are non-convertible to each other. In relativistic mechanics, however, where particles are moving with very high velocity ($v \rightarrow c$), Einstein established that mass and energy are equivalent and interchangeable (mass $\rightleftharpoons$ energy) via conversion factor of c^2 between energy & mass. It is given by

$$E = mc^2 \quad \text{This is known as relativistic total energy.}$$

Where m is relativistic mass of particle when in motion

$$m(v) = \frac{m_0}{\sqrt{1 - \left(\frac{v}{c}\right)^2}}$$

m_0 is the rest mass of particle

The particle at rest also possesses an energy called rest mass energy and is given by $E_0 = m_0 c^2$. It is regarded as the internal energy stored in the particle. Hence the total relativistic energy is the sum of Kinetic and Rest mass energy.

$$E = k + E_0$$

$$E = k + m_0 c^2$$

The law of conservation of total relativistic energy

In Relativistic mechanics, Einstein's mass-energy relation give rise to combined law of conservation of mass & energy

$$[\text{Total relativistic Energy}]_{\text{before}} = [\text{Total relativistic Energy}]_{\text{after}}$$

Derivation From work energy theorem the kinetic energy of a moving body is equal to the amount of work done by the external force on the body from rest.

$$k = \int_0^v F \cdot dx \quad \text{-----}(1)$$

Where dx is the small displacement produced by the amount of force F acting on a particle of mass m . Force is the rate of change of momentum given by

$$F = \frac{d(mv)}{dt}$$

In relativistic mechanics, mass is a function of velocity, therefore

$$F = m \frac{dv}{dt} + v \frac{dm}{dt} \quad \text{-----}(2)$$

$$F \cdot dx = \left(m \frac{dv}{dt} + v \frac{dm}{dt} \right) \cdot dx$$

$$F \cdot dx = m \frac{dv}{dt} dx + v \frac{dm}{dt} dx$$

$$F \cdot dx = m \frac{dx}{dt} dv + v \frac{dx}{dt} dm$$

$$F \cdot dx = m v dv + v^2 dm \quad \text{-----}(3)$$

For particle moving with very high velocity ($v \rightarrow c$),

mass varies relativistic as $m = \frac{m_0}{\sqrt{\left(1 - \frac{v^2}{c^2}\right)}}$ -----(4) $m_0 =$ rest mass

$$m^2 (c^2 - v^2) = m_0^2 c^2$$

$$m^2 c^2 - m^2 v^2 = m_0^2 c^2$$

Differentiating above equation

$$c^2 2m dm - (m^2 2v dv + v^2 2m dm) = 0$$

$$c^2 dm - (m v dv + v^2 dm) = 0$$

$$m v dv + v^2 dm = c^2 dm \quad -$$

$$m v dv = c^2 dm - v^2 dm \quad \text{-----}(5)$$

Substituting eq.(5) into eq.(3)

$$F \cdot dx = c^2 dm \quad \text{-----} (6)$$

Substituting eq (6) back into the eq.(1) gives the Total kinetic energy acquired by particle, moving with velocity v

$$k = \int_{m_0}^m c^2 dm$$

$$k = (m - m_0)c^2$$

$$k = \Delta mc^2 \quad \text{-----}(7)$$

This is the relativistic expression for kinetic energy, i.e. the gain in Kinetic Energy when mass increases with velocity.

Relativistic total energy is the sum of Kinetic & Rest mass energy.

$$E = k + E_0 \quad \text{-----}(8)$$

Where E_0 is the rest mass energy regarded as internal energy stored in the particle. $E_0 = m_0 c^2$

$$E = \Delta m c^2 + m_0 c^2$$

$$E = m c^2 - m_0 c^2 + m_0 c^2$$

$$E = m c^2 \quad \text{-----}(9)$$

This is the universal Einstein “Mass-Energy equivalence” relation

Relativistic expression for Kinetic energy $k = E - E_0$

$$k = m c^2 - m_0 c^2$$

$$k = \left[\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} - 1 \right] m_0 c^2$$

$$k = \left[\left(1 - \frac{v^2}{c^2} \right)^{-1/2} - 1 \right] m_0 c^2$$

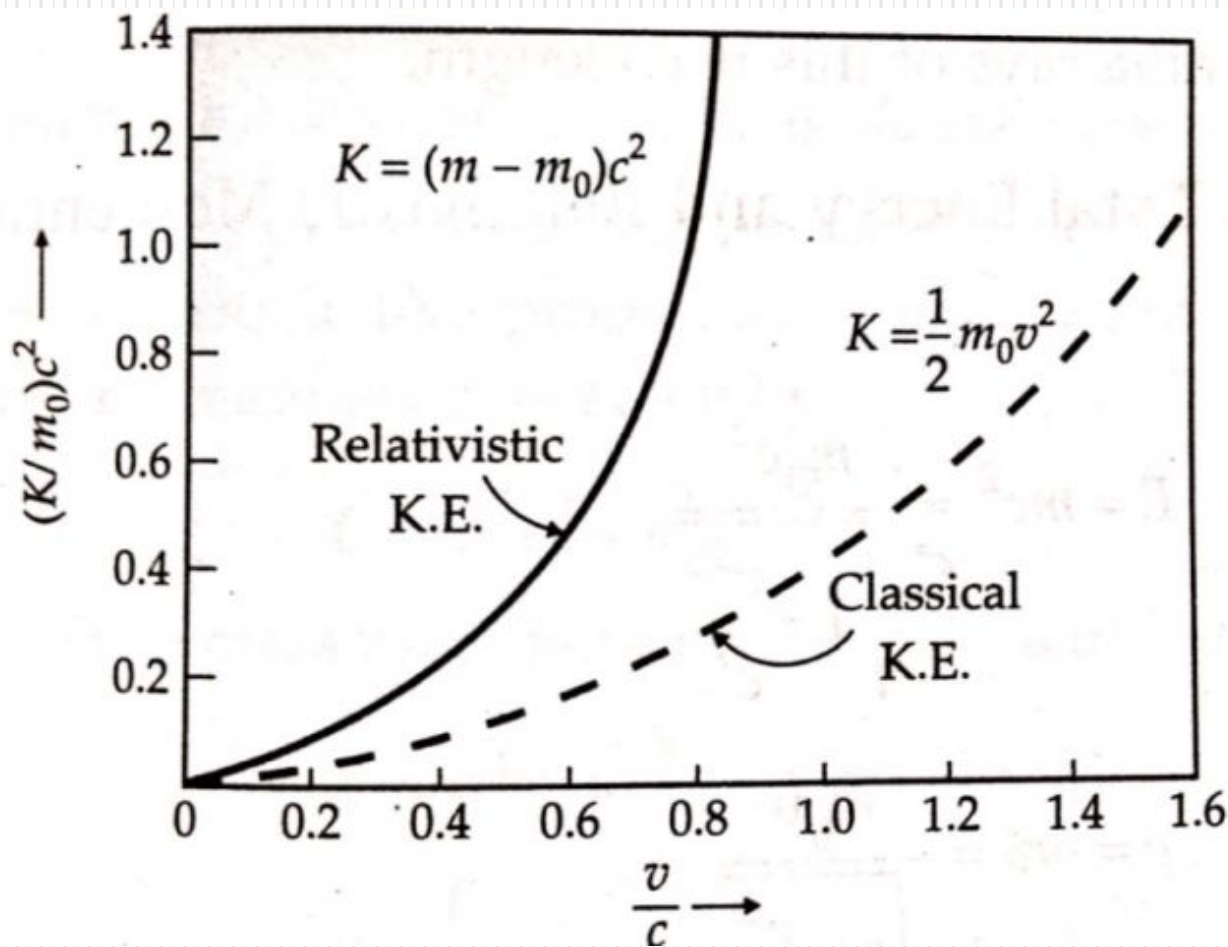
For small velocity ($v \ll c$), the Relativistic expression reduces to the non-relativistic classical expression of Kinetic energy as

$$\frac{v^2}{c^2} \ll 1 \quad \therefore \text{Expanding first term Binomially as}$$

$$k = m_0 c^2 \left(1 + \frac{v^2}{2c^2} \right) - m_0 c^2$$

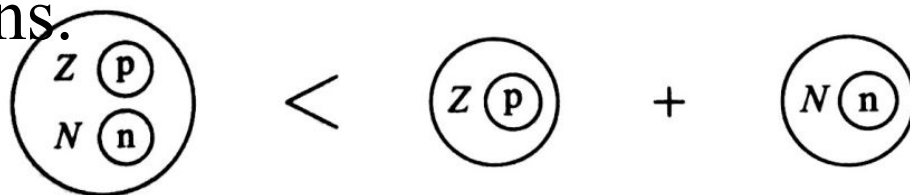
$$k = \frac{1}{2} m_0 v^2 \text{ represents non-relativistic expression of K.E}$$

For small velocity ($v \ll c$), the Relativistic expression for Kinetic energy reduces to the non-relativistic classical expression



Physical significance of mass energy relation

During the formation of a nucleus, sum of the mass of protons & neutrons in a nucleus is greater than the total mass of Nucleons.



$${}^A_Z M_{(p+n)} < Zm_p + nm_n$$

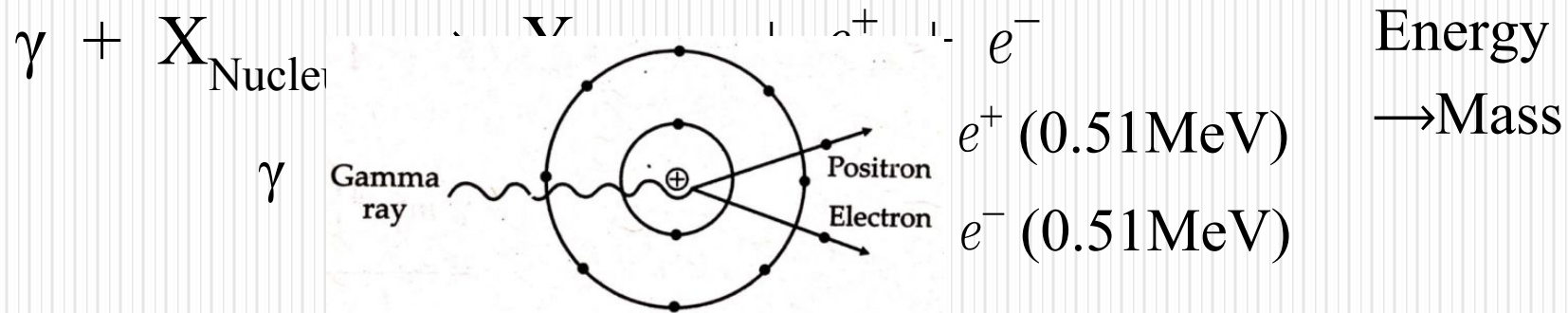
The difference in the mass is known as Mass defect given by

$$\Delta m = (\text{total mass of nucleons}) - (\text{Mass of Nucleus})$$

$$\Delta m = (Zm_p + nm_n) - {}^A_Z M_{(p+n)}$$

Mass defect is converted to energy known as Binding energy, which hold the nucleon together and is given by $E = \Delta m c^2$

Pair production When a photon of energy ≥ 1.02 MeV comes near an atomic nucleus, it disappears & a pair of electron (e^-) and positron (e^+) is created, without violating the conservation laws.



For pair production to happen a photon of at least 1.02 MeV energy is required, since the rest mass energy of electron & positron is 0.51MeV. Any additional photon energy will appear as K.E of electron, positron & recoil nucleus.

Pair annihilation When electron & positron encounter each other under the influence of their opposite electric charges, they annihilate each other & an equivalent amount of energy is produced in the form of γ -rays photon. Their lost mass is converted into energy of 1.02 MeV, in the form of γ -rays photons.



Mass disappears $\rightarrow$ Energy is released

Note: Both these processes are in accordance with the law of conservation of momentum, charge & mass-energy relation. The process of pair production & pair annihilation also occurs in other particle-antiparticle pairs such as proton-antiproton etc.

Relativistic relation for total energy: $E^2 = p^2 c^2 + m_o^2 c^4$

$$\text{Total energy: } E = \frac{m_o c^2}{\sqrt{\left(1 - \frac{v^2}{c^2}\right)}} \quad \text{-----(1)}$$

$$\text{Momentum: } p = \frac{m_o v}{\sqrt{\left(1 - \frac{v^2}{c^2}\right)}} \quad \text{-----(2)}$$

$$E^2 = \frac{m_o^2 c^4}{\left(1 - \frac{v^2}{c^2}\right)} \quad \&$$

$$p^2 = \frac{m_o^2 v^2}{\left(1 - \frac{v^2}{c^2}\right)}$$

$$p^2 c^2 = \frac{m_o^2 v^2 c^2}{\left(1 - \frac{v^2}{c^2}\right)}$$

Now subtracting $p^2 c^2$ from E^2

$$E^2 - p^2 c^2 = \frac{m_0^2 c^4 - m_0^2 v^2 c^2}{\left(1 - \frac{v^2}{c^2}\right)}$$

$$E^2 - p^2 c^2 = \frac{m_0^2 c^4 \left(1 - \frac{v^2}{c^2}\right)}{\left(1 - \frac{v^2}{c^2}\right)}$$

$$E^2 - p^2 c^2 = m_0^2 c^4$$

$$E^2 = p^2 c^2 + m_0^2 c^4 \quad \text{-----}(3)$$

This is the Relativistic relation for total energy that works well for

‘mass particle’ as well as ‘massless particle’